Fake News Classification

The impact of word embeddings and feature selection methods

Sotirios Vantarakis
MSc Data Science
Department of Computer Science
University of Nicosia
vantarakis.sotiris@gmail.com

ABSTRACT

The rise of misinformation in mainstream media has become a pressing topic among scientific communities. The explosive growth of online news content has made it increasingly difficult to distinguish between real and fake news (Fayaz et al., 2022). With the rise of misinformation fake news classification has become a major focus on research. Fake news classification utilizes sophisticated computational techniques to identify false or misleading information from digital content, primarily online news and social media posts. Various techniques are used in this field, including natural language processing (NLP), machine learning, and deep learning, to detect patterns, linguistic markers, or contextual signs associated with fake news. The primary objective is to develop automated systems capable of accurately distinguishing deceptive information. All these models use as input word embedding methods of the news statements and other metadata of a dataset. Another critical challenge is optimal feature selection, where researchers use selection techniques to identify the most predictive features, enhancing model performance beyond traditional machine learning techniques (Fayaz et al., 2022). The purpose of this project is to evaluate the impact of different word embeddings combined with feature selection methods on machine learning and get some insights of which implementation works best. The overall mean accuracy for binary classification was 0.61 while mean precision 0.45, mean recall = 0.46 and mean F1 scire = 0.45, with the best performing algorithm being Logistic Regression with Word2Vec using Skip-Gram architecture, and Chi2 feature selection, with AUC of ROC 0.65. For Random Forest the TF-IDF without feature selection scored AUC of ROC 0.67.

KEYWORDS

NLP, Fake news classification, word embeddings, feature selection

1 Introduction

In the age of information, the rapid spread of news and its consumption via online platforms have become a rising problem. Fake news refers to false or misleading information presented as credible news or statements, often with the intent to deceive readers for political, financial, or ideological gain. This has prompted researchers to explore machine learning and artificial intelligence solutions.

Nowadays the golden standard at research is using neural networks methods to identify and analyze text in depth and classify fake news, with new methods being researched, such as using the BERT model with an LSTM layer, showing promise (Rai et al., 2022). However, machine learning, particularly supervised learning techniques, has proven effective in analyzing large datasets, combining word embeddings and complementary features to classify news as fake or real.

The success of these models depends on several factors. First things first, data quality is of the utmost importance as dataset must be reliable with accurate labels and comprehensive metadata. The process of transforming raw textual data into meaningful numerical representations (feature engineering methods) play a crucial part. Lastly, the selection of algorithms, choosing robust models capable of generalizing well to unseen data and fine tune their parameters.

Studies using neural networks (deep learning methods) on fake news classification, use word embeddings as the only input to analyze the dataset. Thus, the method of choice regarding word embedding has a huge impact on the model. Understanding this, the idea of this project was to try and find which word embedding method works best as a feature for machine learning algorithms to classify fake news.

The present project adopts a multi-stage approach to address the problem of fake news classification. This involves the preprocessing of textual data to remove noise and standardize input features, extraction of features using several word embeddings. Training and evaluating two machine learning models, Random Forest, Logistic Regression.

2 Materials

2.1 Dataset

The dataset selected for the purpose of this project is the LIAR dataset, a benchmark dataset, one of the most reliable and used in research, for political fact-checking to study fake news detection. The dataset, introduced by William Yang Wang in 2017, is an open-source benchmark dataset for fake news detection. It comprises 12,836 manually labeled short statements, collected over a decade from PolitiFact.com, a fact-checking website. includes rich metadata such as speaker identity, political affiliation, and context. This dataset allows for a detailed analysis of patterns and trends in deceptive news content.

This benchmark dataset contains the following columns ID, label, statement, subject, speaker, speaker's job title, state info, party affiliation, total credit history, barely true counts, false counts, half true counts, mostly true counts, pants on fire counts, and the context. It has a 6-label system as meter in decreasing level of truthfulness. TRUE: The statement is accurate and there's nothing significant missing. MOSTLY TRUE: The statement is accurate but needs clarification or additional information. HALF TRUE: The statement is partially accurate but leaves out important details or takes things out of context. MOSTLY FALSE: The statement contains an element of truth but ignores critical facts that would give a different impression. FALSE: The statement is not accurate. PANTS ON FIRE: The statement is not accurate and makes a ridiculous claim.

The dataset includes both textual data (statements) and metadata, offering a comprehensive framework for fake news classification. Based on PolitiFact, for binary classification we grouped together True and Mostly-True as TRUE and the rest are FALSE.

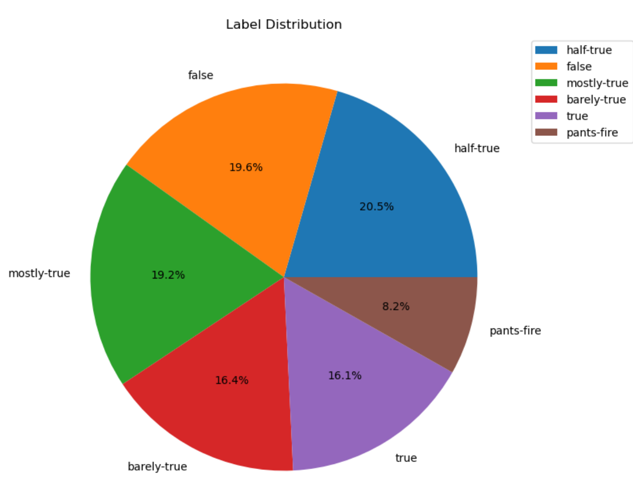

Figure 1: The Label distribution of the truthfulness system used

3 Methods

3.1 Preprocessing

Preprocessing is a critical step in preparing raw data for machine learning tasks. It involves cleaning, transforming, and organizing the dataset to enhance its quality and ensure compatibility with predictive models. Preprocessing not only reduces noise in the data but also ensures that machine learning models can effectively learn patterns and relationships. By removing noise, and addressing data imbalances, preprocessing lays the foundation for accurate and robust fake news detection systems.

First things first, the statements were cleaned from characters such as punctuation. Punctuation, special characters, urls, emails and numbers were removed. Then all text was converted to lowercase to standardize the dataset and reduce feature dimensionality, while any extra whitespaces were stripped. As a next step the stop words were removed. Stopwords are considered frequently occurring but non-informative words (e.g., "the," "is," "and"). To do so predefined lists from NLTK library was used to filter out these terms.

The next steps was tokenization, to split the text in words, NLTK tokenizer was used. Then stemming and lemmatization took place to reduce words and their variation to their root form (running to run), NLTK's Porter Stemmer was used, and normalize words to their base or dictionary form (better to good) retaining semantic meaning, NLTK's WordNet Lemmatizer. Finally for words like 'soooon', which are typos or as ways to gain attention, we reduce them to 'son'.

3.2 Feature extraction

The next step after preprocessing is feature extraction. Feature Extraction involves transforming text and data into numerical features for machine learning models to interpret. For the purposes of this project different word embedding methods on the text were used which will be analyzed.

Word embeddings are numerical representations of words in a continuous vector space, capturing their semantic meaning based on their usage, frequency and context within a corpus. These embeddings serve as the foundation for many natural language processing (NLP) tasks. Several techniques are used to generate word embeddings, each with unique characteristics and advantages.

The following section explores the word embedding methodologies used in this project and provides specific insights into Bag of Words (BoW), Term Frequency-Inverse Document Frequency (TF-IDF), Word2Vec and GloVe.

The Bag of Words model is the simplest method for transforming text into numerical form. It represents each document as a vector of word frequencies, disregarding grammar, word order, or semantics. The approach is straightforward, vocabulary is created from all unique words in corpus and each document is represented as a vector, where each dimension corresponds to a word in the vocabulary.

TF-IDF is an extension of BoW that assigns weights to words based on their importance and rarity within a document relative to the entire corpus. This reduces the impact of common words and highlights informative terms. Term frequency (TF) measures how often a word appears in a document. This method Emphasizes distinctive words, improving relevance for classification tasks, while mitigating the effects of overly common words like "the" or "and."

Word2Vec is a neural network-based model that generates dense, low-dimensional word embeddings. Developed by Mikolov et al., it captures semantic and syntactic relationships through two architectures: Skip-Gram and Continuous Bag of Words (CBOW). Skip-Gram Predicts the surrounding words (context) given a target word. It works by maximizing the probability of context words, given a target word. The other approach CBOW, predicts the target word given its context words. Both models learn embeddings by maximizing the probability of co-occurrence.

GloVe is a matrix factorization-based embedding technique that uses co-occurrence statistics to generate word vectors. It was developed by researchers at Stanford University to address limitations in Word2Vec. It records how often word pairs appear together in the corpus. The method works by constructing a word co-occurrence matrix from the corpus. Then factorizes this matrix to create dense word embeddings and optimizes the loss function to ensure that word pairs with similar contexts have similar embeddings.

3.3 Feature Selection methods

In machine learning, high-dimensional data can lead to challenges such as increased computational cost, model overfitting, difficulties in visualization and even lower model predictions. To address these issues, dimensionality reduction and feature selection techniques are applied to transform or select a subset of features from the original dataset while retaining as much information as possible. Below, we explore five key methods used in this project namely, Principal Component Analysis (PCA), Mutual information gain, Chi-Square Test, and Tree-Based Feature Selection, explaining how they work and their role in improving model performance.

Principal Component Analysis (PCA) is a widely used dimensionality reduction technique that transforms a dataset into a new coordinate system, where the axes (called principal components) are linear combinations of the original features. These principal components are ordered based on the variance they capture, with the first few components retaining most of the data's variability. The data is projected onto the top k principal components to reduce dimensionality while preserving significant variability.

Mutual information is a measure of the dependency between two variables. In feature selection, it evaluates the amount of information a feature provides about the target variable. It is particularly useful for selecting features that have a nonlinear relationship with the target. Mutual information is effective for identifying non-linear dependencies and selecting features that strongly influence the target. It works well with both numerical and categorical data but can be computationally expensive for large datasets.

The Chi-Square Test is a statistical method used for feature selection, particularly in classification tasks with categorical data. It evaluates the independence between a feature and the target variable by testing whether the observed distribution deviates significantly from the expected distribution. It is Simple and effective for categorical data. It helps identify features with strong statistical relationships to the target variable.

Tree-based methods, such as Decision Trees, can be used for feature selection by measuring the importance of features based on their contribution to the predictive performance of the model. During training, tree-based algorithms evaluate the importance of each feature by assessing how much it reduces the impurity (e.g., Gini impurity or entropy) at each split. Features that contribute the most to reducing impurity are ranked higher in importance.

Each method offers unique benefits and challenges, for example, PCA is ideal for high-dimensional data with numerical features, while Chi-Square excels with categorical features and label associations. Mutual Information identifies nonlinear dependencies, and tree-based methods provide interpretable importance rankings that work well for mixed data types.

3.4 Machine Learning algorithms

Machine learning algorithms and their choice, play a crucial role in the classification tasks of fake news detection. Searching through research papers made on this topic, two widely used algorithms are Logistic Regression and Random Forest. Both algorithms have distinct approaches and advantages, making them effective tools for handling complex datasets. Below, we delve into their working principles, key concepts, and how they contribute to classification tasks.

Logistic Regression is among the most frequently applied algorithms in classification. By adding a sigmoid function at the end, logistic regression yields a logistic graph that is restricted between 0 and 1 values. It predicts the probability that a given input belongs to a particular class, making it especially useful for binary classification problems.

Random Forest is an ensemble learning method that simply constructs multiple decision trees during training and combines their outputs to produce more accurate results and reduce overfitting. It is widely used for both classification and regression tasks due to its robustness and interpretability. A decision tree is a flowchart-like structure where each internal node represents a test on a feature, each branch represents an outcome, and each leaf node represents a class label or output value. Trees are constructed by recursively splitting the dataset based on features that maximize a splitting criterion, such as Gini Impurity or Entropy.

3.4 Evaluation Metrics

Evaluating machine learning models is crucial to assess their performance and ensure their predictions are reliable. For classification tasks, metrics such as accuracy, precision, recall, and F1-score are commonly used. These metrics offer complementary insights into a model's effectiveness, particularly in imbalanced datasets and when misclassification matters, where relying solely on accuracy can be misleading.

Accuracy measures the proportion of correctly classified instances, both true positives and true negatives, out of the total instances in the dataset. It provides a general sense of how well the model performs overall.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision focuses on the proportion of true positive predictions among all instances classified as positive.

$$Precision = \frac{TP}{TP + FP}$$

High precision means the model has fewer false positives, making it reliable in scenarios where false positives are costly.

Recall, also known as sensitivity, measures the proportion of actual positive instances that the model correctly identifies.

$$Recall = \frac{TP}{TP + FN}$$

High recall indicates that the model correctly identifies most of the positive instances, minimizing false negatives. Recall is crucial when missing positive instances is costly.

The F1-score is the harmonic mean of precision and recall, providing a single metric that balances both. It is particularly useful when there is a trade-off between precision and recall, or when the dataset is imbalanced.

$$F1 - Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

An F1-score ranges from 0 to 1, with 1 being the best possible score.

ROC AUC is a metric used to evaluate the performance of classification models, particularly for binary classification tasks. It plots the True Positive Rate (Sensitivity) against the False Positive Rate (1 - Specificity) at various threshold levels. The area under the ROC curve summarizes the model's ability to distinguish between classes.

4 Experiment

For the experiment's implementation different classification algorithms were used with different feature extraction methods for the statements text.

The method used was Python 3.12.3 for the experiment. The dataset was unified to perform k-fold cross validation, a common practice in machine learning. 5-fold cross validation was used because too few folds would result in bias and unreliable performance, while using too many folds leads to very long training times without significant improvement. Thus, 5-fold provides a good balance between bias and high training times, offering a stable and reliable estimate of model performance.

For feature extraction the word embedding methos were used were, Bag of Words, TF-IDF, Glove and word2vec using both architectures. BoW and TF-IDF were limited to 5000 features and used unigrams, bigrams and trigrams. Word2vec vector size was set to 300 dimensions. High dimensions capture better and richer semantic relationships between words, while higher leads to overfitting without significant performance gains with both SkipGram and CBOW methods used. For Glove the 300 dimensions were used.

Furthermore, as complementary features, to strengthen the performance, the negative and positive word count in the statements was used, utilizing the NLTK opinion lexicon. Some of the metadata of the LIAR dataset were also used. The venue, speaker's job title and subject of the statement were firstly grouped in smaller groups to have greater meaning as features (e.g. to generalize otherwise it would be like ID and be a bad feature) and then encoded to number, making them able to be used by the algorithms as features.

For the algorithms HalvingGridSearchCV was used to try and specify some best performing parameters. The dataset was a bit imbalanced, so the scoring metric selected was not accuracy but AUC of the ROC curve as it is a better metric especially for imbalance datasets. Lastly, an important challenge in the LIAR dataset is addressing imbalanced label distributions. This project used binary classification where there was a small imbalance to label (65% False and 35% True approximately). To mitigate this, Synthetic Minority Oversampling Technique (SMOTE) was employed to generate synthetic samples for underrepresented labels, and the class weight was set to balanced in the parameters of the algorithms. This way the algorithm penalizes the miclassification of minority class.

Lastly feature selection methods were used to reduce the features to the most important ones. These were PCA with variance set to 95%, Mutal information gain, chi-2 were set to select the k =100 best features and tree based feature selection using random fore classifier.

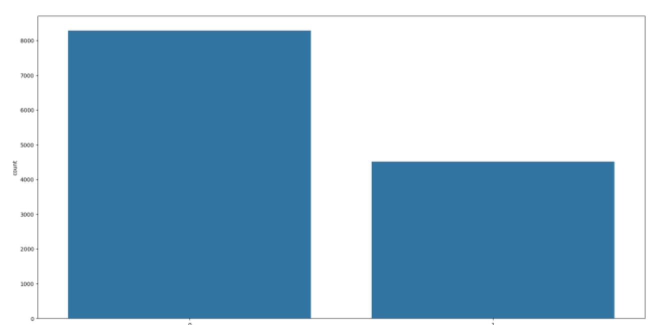

Figure 2: Distribution of the binary label classes used

5 Results

The accuracy, precision, recall, and F1-score outcomes of each algorithm for each word embedding and each feature selection method were putted in tables. Overall the best performance was when no feature selection method was applied, though not much difference (less than 0,1 in each metric). A plot for each classifier is shown below.

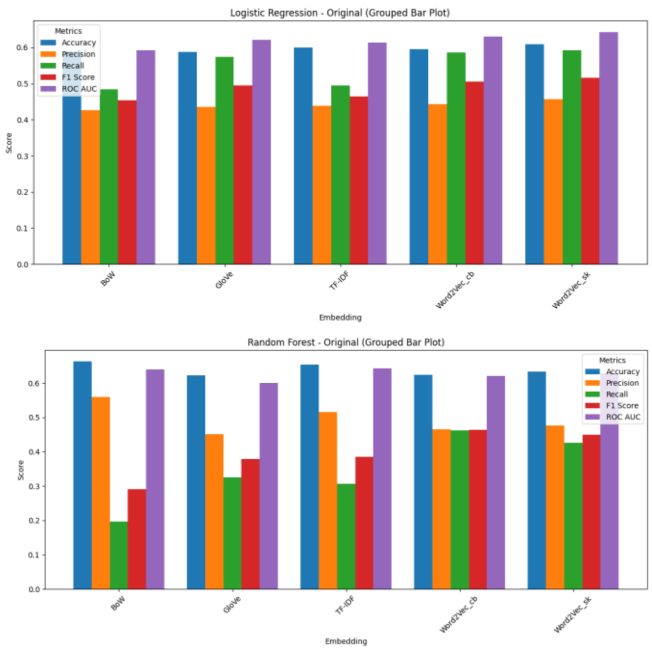

Figure 3: Logistic Regression and Random Forest for no feature selection for each embedding method

The mean confusion matrix of the whole experiment is shown below.

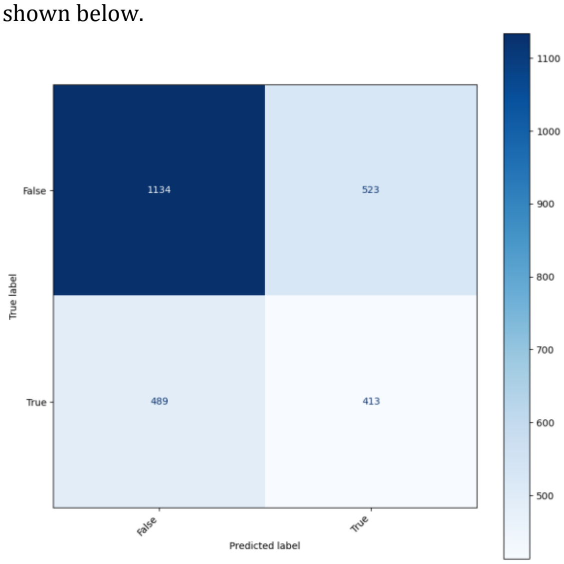

Figure 4: Mean Confusion matrix for the total experiment (not the best performing one)

6 Discussion

The performance of each classifier, embedding and feature selection method pairing was evaluated. It is evident that Logistic Regression performed better overall. Even though Random Forest showed the best accuracy metric, the recall and precision were very low, in contrast to logistic regression whose metrics were all above average.

For logistic regression Word2Vec and specifically Skip-Gram method had the best results with an AUC ROC of 0.642, Recall of 0.591 and Precision 0.457, followed by GloVe with AUC ROC 0.621.

Random Forest showed the best results with TF-IDF and Bag of words but even though the accuracy was 0.654 and 0.662, the recall was 0.307 and 0. 197 respectively, showing poorer overall performance.

Regarding feature selection methods the difference in metrics between the best performing models is depicted for both Logistic Regression and Random Forest, for the different feature selection methods. Nearly every metric has decreased with few exceptions.

	Logistic Regression					Random Forest				
	Accuracy	Precision	Recall	F1 Score	ROC AUC	Accuracy	Precision	Recall	F1 Score	ROC AUC
Non	0	0	0	0	0	0	0	0	0	0
PCA	-0.038	-0.043	-0.069	-0.054	-0.080	-0.099	-0.15562	0.029832	-0.03677	-0.12976
MutaulInfo	-0.004	-0.005	-0.011	-0.007	-0.006	-0.061	-0.112	0.019	-0.024	-0.086
Chi-Square	-0.006	-0.00604	-0.00335	-0.00516	-0.00459	-0.054	-0.09909	0.029232	-0.01486	-0.07351
Tree-Based	-0.006	-0.00592	-0.00205	-0.00459	-0.00515	-0.055	-0.07439	0.207951	0.090444	-0.02261

Figure 5: Difference between feature selection methods for the best performing model.

When using the BoW and TF-IDF embedding methods, each unique word is used as a predictor in our classification methods. However, the document-term matrix is highly sparse with these methods, leading to convergence issues for many classification methods.

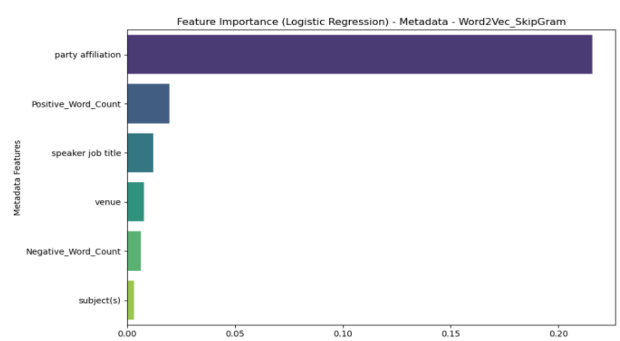

Figure 6: Best performing complementary features for Logistic Regression using Word2Vec Skip-Gram

Regarding the metadata the most important feature was party affiliation followed by the subject in most cases. For Logistic Regression utilizing word2vec, the best feature was by far party affiliation, which actually might raise ethical problems.

The overall mean accuracy for binary classification was 0.61 while mean precision 0.45, mean recall = 0.46 and mean F1 score = 0.45

7 Conclusion and Future Work

This study evaluated various word embedding techniques and feature selection methods for fake news classification, using the LIAR dataset as a benchmark. Among the tested approaches, Logistic Regression paired with Word2Vec (Skip-Gram architecture) and Chi-Square feature selection yielded the best overall performance, achieving an AUC ROC of 0.64. Random Forest also performed well with TF-IDF embeddings but demonstrated lower recall and precision, highlighting a trade-off in algorithm selection based on specific metrics. The best overall feature was Word2Vec skip-gram variation, while for the metadata it was the party affiliation.

While the results indicate promising directions for detecting fake news, challenges such as data imbalance and feature sparsity remain. Feature selection methods showed marginal benefits in some cases, though overall performance was often better without their application showcasing a possible loss of information.

Synthesizing all these results, it is clear that the topic is very multidimensional and proved challenging. A lot more can be implemented as future work. In this project gridsearch was not used as it is computational and time expensive so as a future work it could be used to find the best performing parameters of our algorithms. Also, more algorithms should be used to identify which performs the best, for example SVM was not used, though it is considered one of the best performing algorithms for classification. In addition, the dataset imbalance should be researched more thoroughly. SMOTE method that was used, created synthetic data by interpolating existing samples to fix the imbalance, which might lead to overfitting problems, noise amplification or even class overlap. Furthermore, the best performing features and feature selection methods could be used as input in a deep learning neural network method, for richer analysis. Finaly, not only could more word embedding methods be researched but also used for multilabel classification for which the dataset could be used for.

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